SECURITY OPERATIONS / HOME LAB

Simulated incident analysis

Authentication failures followed by a successful login

SIMULATION — No real host, user, network packet or account was monitored. The IP 203.0.113.66 belongs to a range reserved for documentation and does not identify a real source.

1. Executive summary

The **compromised** scenario, seed **2026**, generated 16 synthetic events. Five login failures for **admin** came from the same IP between 12:30 and 12:34 UTC; a success for the same account and source followed at 12:35 UTC. The detector produced a **medium** alert for repeated failures and a **high** alert for the correlation with the success. The high alert was recorded as **investigating**; the medium one remains **open**.

Exercise conclusion: the events meet the implemented rules. This does not prove a real compromise; original sources and context must be validated before classifying an incident.

2. Scope and reproduction

Item	Value	What it represents
Scenario / seed	compromised / 2026	Reproducible simulator input.
Events	16	10 context events and 6 events in the test sequence.
Host	auth-server-01	Fictitious name included by the generator.
Run ID	ac958c2bb9d5494b921fd16a81719abc	Run recorded in the local SQLite database.
Response	None	No block or automated action was executed.

Reproduction without changing the main database:

```
./.venv/bin/python -m soc_lab.simulator --scenario compromised --seed 2026 --output /tmp/soc-lab-compromised-2026.jsonl
./.venv/bin/python -m soc_lab.cli --input /tmp/soc-lab-compromised-2026.jsonl --db /tmp/soc-lab-analysis.sqlite
```

3. Observed timeline

Time (UTC)	Synthetic event	Interpretation
12:30	Failure for admin from 203.0.113.66	First failure.
12:31–12:33	Three more failures for the same account and source	Count rises to four.
12:34	Fifth failure for the same account and source	The repeated-failures rule reaches the 5/10 min threshold.
12:35	Success for admin from the same IP	The correlation rule finds a success within 5 min.

4. Field meanings

Field	Example	Meaning / limits
timestamp	12:30–12:35 UTC	Fictitious event time; UTC makes it easy to order sources.
event_type	authentication	Classifies the record; it is not a packet capture.
source_ip	203.0.113.66	Synthetic IP. It does not identify a person or a real network.
username	admin	Fictitious target account; it does not reveal who tried to authenticate.
outcome	failure / success	Result produced by the simulator, not confirmed by a real IdP.
hostname	auth-server-01	Fictitious host name assigned to the events.
event_id	evt-0011…evt-0016	Sequential identifier to locate records in the set.
scenario	compromised	Test label; it is not an incident verdict.

5. Triggered rules

multiple_failed_logins — medium. Groups failures by IP and checks for at least five in a sliding ten-minute window. This rule does not need to know the user; that is why it is not the rule that states that admin authenticated successfully.

successful_login_after_failures — high. Correlates by the IP + user pair. It found five failures for admin between 12:30 and 12:34 and a success at 12:35. The one-minute gap is within the configured five-minute window, so the severity was raised.

These are two alerts about the same events, not two independent occurrences. A more complete SIEM could group both into a single case to reduce duplication.

6. Triage reasoning

Category	Analysis
Observation	Five failures and a subsequent success share the same IP and user in the generated set.
Hypothesis	If these were real logs, the sequence would justify investigating credential use after repeated attempts.
Not demonstrated	There is no evidence of a discovered password, real access, operator identity, MFA or post-login activity.
Benign alternatives	Typo before the success, authorized testing, support activity or a shared intermediate address.
Confidence	High that the rule matches the synthetic events; no conclusion about a real environment.

7. Recorded decision and next steps

The high alert, ID **22d19b8ead764333bb959ce0bde6812f**, is **investigating**. The stored note explicitly marks the simulation and recommends checking original logs before concluding a compromise. The medium alert, ID **c53808f573ce4dec8578f0303dc0605f**, remains **open**.

• Confirm IP, host, user and method in the original logs of the system or identity provider.

• Check whether the success and the device/session were expected and review the activity that followed.

• Check MFA, source and account history to compare with normal behavior.

• If real evidence confirms risk, follow the organization's authorized procedure to protect the account and revoke sessions.

8. Limitations and conclusion

The lab analyzed synthetic JSONL only. It did not read Windows/Linux logs, did not observe traffic, did not analyze TCP/UDP and did not run any automated response. The **investigating** status is part of the exercise. Never block the documentation IP based on this result; in a real case, validate the original evidence and follow the authorized process.